

#BuildingResilience — WhatsApp Platform Roadmap (Deliverable 2)												
Feature ID	Component	Feature	Description	Priority	Phase	Dependencies	Owner	Status	Target Week	Complexity	Notes	
COMP A: TWILIO / META SETUP												
WA-A01	Twilio / Meta	Twilio account creation	Create Twilio account and upgrade to paid plan	P0	Phase 1	None	BOMALAB	Done	W1	Low	Account SID and Auth Token secured	
WA-A02	Twilio / Meta	UK regulatory bundle	Create individual regulatory bundle for UK mobile number (Ofcom requirement) — submit ID and proof of address	P0	Phase 1	WA-A01	BOMALAB	Done	W1	Low	Approved same day	
WA-A03	Twilio / Meta	UK mobile number purchase	Purchase UK mobile number on Twilio, assign approved regulatory bundle	P0	Phase 1	WA-A02	BOMALAB	Done	W1	Low	Number purchased and active	
WA-A04	Twilio / Meta	Meta Business Portfolio	Create new Meta Business Portfolio under CHOMEI COMMERZ LTD during Self Sign-up flow	P0	Phase 1	WA-A03	BOMALAB	Done	W1	Low	Created via Twilio Self Sign-up	
WA-A05	Twilio / Meta	Meta Business Verification	Submit business verification with Companies House documentation, website URL	P0	Phase 1	WA-A04, company website	BOMALAB	Done	W1	Low	Approved same day	
WA-A06	Twilio / Meta	WhatsApp sender registration	Register UK number as WhatsApp sender via Twilio Self Sign-up, create WABA	P0	Phase 1	WA-A03, WA-A04	BOMALAB	Done	W1	Low	Submitted, awaiting Meta review (up to 24h)	
WA-A07	Twilio / Meta	Sender display name	Configure WhatsApp sender display name (e.g. 'BuildingResilience' or 'BOMALAB')	P0	Phase 1	WA-A06	BOMALAB + OC	Not started	W1	Low	Need to agree name with OC, Meta reviews name	
WA-A08	Twilio / Meta	Business profile setup	Configure business profile: description, website, address, category, profile photo	P1	Phase 1	WA-A06	BOMALAB + OC	Not started	W1	Low	Visible to participants in business info screen	
WA-A09	Twilio / Meta	Twilio sandbox setup	Configure Twilio sandbox for WhatsApp development and testing	P0	Phase 1	WA-A01	BOMALAB	Not started	W1	Low	Enables immediate development while awaiting sender approval	
WA-A10	Twilio / Meta	Message templates	Create and submit message templates for broadcasts: welcome, event reminder, session announcement, post-event follow-up (FR + EN)	P0	Phase 1	WA-A06	BOMALAB + OC	Not started	W2	Medium	Templates require Meta approval, submit early	
WA-A11	Twilio / Meta	Company website (chomei.store)	Deploy landing page for CHOMEI COMMERZ LTD to support Meta business verification	P0	Phase 1	None	BOMALAB	Done	W1	Low	Live at chomei.vercel.app, DNS pending for chomei.store	
COMP B: FASTAPI BACKEND												
WA-B01	Backend	FastAPI project scaffold	Project structure, dependencies, environment config, Vercel/Railway deployment config	P0	Phase 1	None	BOMALAB	Not started	W1	Low	Python 3.10+, FastAPI, uvicorn	
WA-B02	Backend	Twilio webhook handler	POST endpoint that receives incoming WhatsApp messages from Twilio, parses sender number, message body, media	P0	Phase 1	WA-B01, WA-A09	BOMALAB	Not started	W1	Medium	Reference: World Bank WhatsApp-RAG-Example main.py	
WA-B03	Backend	Twilio response handler	Function to send WhatsApp replies back through Twilio API, handle message formatting and character limits	P0	Phase 1	WA-B02	BOMALAB	Not started	W1	Medium		
WA-B04	Backend	Claude API integration	Connect to Anthropic Claude API, construct prompts with system prompt + knowledge base + user message, parse responses	P0	Phase 1	WA-B02	BOMALAB	Not started	W1	Medium	Direct context injection approach (not RAG)	
WA-B05	Backend	Language detection & routing	Detect incoming message language, configure Claude to respond in same language, default to FR/EN if unsupported	P0	Phase 1	WA-B04	BOMALAB	Not started	W2	Low	Handled in Claude system prompt	
WA-B06	Backend	Escalation logic	Detect when Claude cannot answer (out of knowledge base scope), offer to connect user with human operator, flag for follow-up	P1	Phase 1	WA-B04	BOMALAB	Not started	W2	Medium	Define escalation rules with OC	
WA-B07	Backend	Supabase conversation logging	Log every conversation (sender hash, message, response, timestamp, language) to Supabase for analytics and debugging	P0	Phase 1	WA-B02, Supabase	BOMALAB	Not started	W1	Medium	Anonymise sender numbers per RGD	
WA-B08	Backend	Rate limiting	Prevent abuse: limit messages per user per time window, handle Twilio rate limits gracefully	P1	Phase 1	WA-B02	BOMALAB	Not started	W2	Low		
WA-B09	Backend	Error handling & resilience	Graceful handling of Claude API timeouts, Twilio errors, malformed messages, retry logic	P0	Phase 1	WA-B02, WA-B04	BOMALAB	Not started	W2	Medium	Critical for event-day reliability	
WA-B10	Backend	Health check endpoint	GET /health endpoint for monitoring, uptime checks, deployment verification	P1	Phase 1	WA-B01	BOMALAB	Not started	W1	Low		
WA-B11	Backend	Production deployment	Deploy FastAPI to Vercel serverless functions or Railway, configure production webhook URL in Twilio	P0	Phase 1	WA-B01 to WA-B09	BOMALAB	Not started	W3	Medium	Swap sandbox webhook URL for production URL	
WA-B12	Backend	Broadcast messaging endpoint	Admin-authenticated endpoint to send template messages to all registered participants or specific groups	P1	Phase 1	WA-A10, WA-B01	BOMALAB	Not started	W3	Medium	Used for #Announces channel functionality	
COMP C: KNOWLEDGE BASE												
WA-C01	Knowledge Base	System prompt design	Design Claude system prompt: persona, tone, scope boundaries, language instructions, escalation rules	P0	Phase 1	None	BOMALAB	Not started	W1	Medium	Defines the assistant's personality and behaviour	
WA-C02	Knowledge Base	Campaign content integration	Structure OIAFLAD background, #BuildingResilience objectives, UNGA launch context into knowledge base format	P0	Phase 1	WA-C01	BOMALAB + OC	Not started	W2	Low	Draft from concept note, OC to validate	
WA-C03	Knowledge Base	Gabon national adaptation	CAP 241 alignment, healthcare continuity, social vulnerability, community resilience context	P0	Phase 1	WA-C01	OC	Not started	W2	Low	BLOCKED until OC provides Gabon-specific content	
WA-C04	Knowledge Base	Event programme & logistics	Session schedule, venue details, transport info, practical FAQ — structured for AI retrieval	P0	Phase 1	WA-C01	OC	Not started	W2	Low	BLOCKED until OC delivers programme	
WA-C05	Knowledge Base	Speaker & partner information	Speaker bios, panel compositions, partner descriptions for the assistant to reference	P1	Phase 1	WA-C01	OC	Not started	W2	Low	BLOCKED until OC delivers speaker data	
WA-C06	Knowledge Base	FAQ compilation	Compile frequently asked questions: registration process, dress code, languages, accessibility, contacts	P0	Phase 1	WA-C01	OC + BOMALAB	Not started	W2	Low	Anticipate common participant questions	
WA-C07	Knowledge Base	Bilingual content (FR + EN)	Ensure all knowledge base content is available in French and English, test responses in both languages	P0	Phase 1	WA-C02 to WA-C06	BOMALAB + OC	Not started	W2	Medium	Portuguese responses handled by Claude natively	
WA-C08	Knowledge Base	Knowledge base update process	Document the process for OC to request knowledge base updates (new info, corrections) during and after event	P1	Phase 1	WA-C01	BOMALAB	Not started	W3	Low	Simple: OC sends update, BOMALAB edits system prompt	
WA-C09	Knowledge Base	Phase 2: RAG migration	If knowledge base outgrows system prompt, migrate to RAG pipeline with Supabase pgvector embeddings	P2	Phase 2	WA-C01 to WA-C07	BOMALAB	Not started	Phase 2	High	Only if content volume exceeds Claude context window	
COMP D: COMMUNITY GROUPS												
WA-D01	Community	WhatsApp Community creation	Create the official WhatsApp Community under the registered business number	P0	Phase 1	WA-A06	BOMALAB	Not started	W2	Low	Requires approved sender number	
WA-D02	Community	#Announces channel	Create broadcast-only announcements channel for official communications	P0	Phase 1	WA-D01	BOMALAB	Not started	W2	Low	Read-only for participants	
WA-D03	Community	#Programme channel	Create programme/agenda channel with session links and sondage integration	P0	Phase 1	WA-D01	BOMALAB	Not started	W2	Low		
WA-D04	Community	#Réseautage channel	Create networking channel for peer connections and introductions	P1	Phase 1	WA-D01	BOMALAB	Not started	W2	Low		
WA-D05	Community	#Exposition channel	Create channel for showcasing women's initiatives and community projects	P1	Phase 1	WA-D01	BOMALAB	Not started	W2	Low		
WA-D06	Community	#Délégation-VIP channel	Create restricted VIP channel for First Ladies and ministerial delegations	P0	Phase 1	WA-D01	BOMALAB + OC	Not started	W2	Low	Invitation-only, tighter access controls	
WA-D07	Community	Invitation flow	Define and test the process for inviting participants to the Community via registration data	P0	Phase 1	WA-D01, WA-A10	BOMALAB	Not started	W2	Medium	Link invitation to website registration	
WA-D08	Community	Welcome message	Automated welcome message when participants join, with orientation and links	P1	Phase 1	WA-D01, WA-A10	BOMALAB + OC	Not started	W3	Low	Requires approved message template	
WA-D09	Community	Ad-hoc group creation process	Document the process for rapidly creating new groups on demand during the event	P1	Phase 1	WA-D01	BOMALAB	Not started	W3	Low	Key flexibility advantage highlighted in proposal	
COMP E: TESTING & LAUNCH												
WA-E01	Testing	End-to-end sandbox testing	Test full flow: user sends WhatsApp message → Twilio → FastAPI → Claude → response via Twilio → user receives reply	P0	Phase 1	WA-B01 to WA-B04, WA-C01	BOMALAB	Not started	W2	Medium	Test in FR, EN, PT	
WA-E02	Testing	Load testing	Simulate 100+ concurrent messages, measure response times, identify bottlenecks	P0	Phase 1	WA-E01	BOMALAB	Not started	W3	Medium	Critical for event-day with 1,000 participants	
WA-E03	Testing	Edge case testing	Test: empty messages, very long messages, images/audio sent, unsupported languages, rapid-fire messages from same user	P1	Phase 1	WA-E01	BOMALAB	Not started	W3	Medium		
WA-E04	Testing	Escalation flow testing	Test human handoff: user asks out-of-scope question → assistant offers human contact → OC receives notification	P1	Phase 1	WA-B06, WA-E01	BOMALAB + OC	Not started	W3	Low	Agree escalation contact with OC	
WA-E05	Testing	VIP soft launch	Invite organising team and VIP delegation to test the platform, collect feedback	P0	Phase 1	WA-E01 to WA-E04, WA-D01 to WA-D06	BOMALAB + OC	Not started	W3	Low	Final validation before public launch	
WA-E06	Testing	Production webhook cutover	Switch Twilio webhook from sandbox/dev to production FastAPI URL, verify live	P0	Phase 1	WA-B11, WA-A06	BOMALAB	Not started	W3	Low		
WA-E07	Testing	Event-day monitoring plan	Define monitoring: uptime checks, error alerting, response time dashboards, escalation contacts	P0	Phase 1	WA-B10	BOMALAB	Not started	W3	Medium	As per proposal: BOMALAB contactable throughout event	
WA-E08	Testing	Post-event transition plan	Plan for post-event: reduce Claude API usage, archive conversations, transition Community to long-term mode	P2	Phase 1	All	BOMALAB	Not started	W3	Low	Feeds into Phase 2 planning	

Content Dependencies — WhatsApp Platform

Content Item	Source	Needed By	Blocks Features	Status	Notes
Agreed WhatsApp sender display name	OC + BOMALAB	Week 1	WA-A07	Pending	Need OC sign-off — 'BuildingResilience' recommended
Campaign messaging / tagline (FR + EN)	OC	Week 2	WA-C02	Pending	For system prompt persona and welcome messages
OAFLAD background text (FR + EN)	OC	Week 2	WA-C02	Pending	Available on oaflad.org, needs adaptation for AI knowledge base
Gabon national adaptation text (CAP 241)	OC	Week 2	WA-C03	Pending	Specific to Gabonese initiative — not publicly available
Event programme: sessions, times, descriptions	OC	Week 2	WA-C04	Pending	CRITICAL — blocks event logistics in knowledge base
Speaker list: names, titles, bios	OC	Week 2	WA-C05	Pending	For AI to reference when asked about speakers
Partner / organisation descriptions	OC	Week 2	WA-C05	Pending	For AI to reference when asked about partners
Venue and logistics details (address, transport, access)	OC	Week 2	WA-C04	Pending	Practical info participants will ask the assistant
FAQ: anticipated participant questions + answers	OC + BOMALAB	Week 2	WA-C06	Pending	BOMALAB can draft, OC to validate and expand
Broadcast message content (FR + EN)	OC	Week 2	WA-A10	Pending	Content for message templates: welcome, reminders, announcements
VIP delegation list for restricted channel	OC	Week 2	WA-D06	Pending	Names/numbers for Délégation-VIP channel invitation
Escalation contact details	OC	Week 3	WA-B06, WA-E04	Pending	Who receives escalated queries from the AI assistant
Registration data feed for WhatsApp invitations	OC / Website	Week 3	WA-D07	Pending	Depends on website registration being operational
Business profile photo / logo	OC	Week 1	WA-A08	Pending	Profile image visible in WhatsApp business info

Weekly Sprint Plan — WhatsApp Platform

Week	Sprint Goal	Features to Complete	Content Needed From OC	Deliverable / Milestone
W1	Infrastructure + core backend	WA-A01 to WA-A06 (DONE), WA-A07 (display name), WA-A08 (profile), WA-A09 (sandbox), WA-A11 (DONE), WA-B01 (scaffold), WA-B02 (webhook), WA-B03 (response), WA-B04 (Claude), WA-B07 (logging), WA-B10 (health), WA-C01 (system prompt)	Agreed display name, business profile photo	Working AI assistant on sandbox: send a message, get an intelligent response. Conversation logging active.
W2	Knowledge base + community setup	WA-A10 (templates), WA-B05 (language), WA-B06 (escalation), WA-B08 (rate limit), WA-B09 (errors), WA-C02 to WA-C07 (all KB content), WA-D01 to WA-D07 (community + channels), WA-E01 (e2e test)	Campaign text, programme, speakers, venue, FAQ, broadcast content, VIP list	Full knowledge base loaded. Community created with all channels. Message templates submitted. End-to-end tested in FR/EN/PT.
W3	Testing, hardening, soft launch	WA-B11 (production deploy), WA-B12 (broadcast), WA-C08 (update process), WA-D08 (welcome), WA-D09 (ad-hoc process), WA-E02 to WA-E07 (all testing), WA-E08 (post-event plan)	Final corrections, escalation contacts, registration data for invitations	Production deployment live. Load tested. VIP soft launch. Monitoring plan in place. Ready for event day.
Event	Live operations (17 April 2026)	All systems operational. Real-time monitoring. Broadcast announcements via #Annonces. AI assistant handling participant queries. Ad-hoc groups created as needed.	Real-time broadcast content from OC	WhatsApp platform fully operational throughout event. BOMALAB on standby for technical support.
Post	Transition to campaign mode	WA-E08 (transition), WA-C09 (RAG migration assessment). Reduce API usage, archive conversations, assess Phase 2 needs.	Decision on long-term community management approach	Platform transitioned to long-term mode. Analytics report delivered. Phase 2 scope confirmed.

Supabase Schema — WhatsApp Platform

Table	Column	Type	Constraints	Description
wa_conversations				
wa_conversations	id	uuid	PK, default gen_random_uuid()	Conversation ID
wa_conversations	created_at	timestamptz	NOT NULL, default now()	Conversation start timestamp
wa_conversations	sender_hash	text	NOT NULL	SHA-256 hash of sender phone number (RGPD: no raw numbers stored)
wa_conversations	language	text	default 'fr'	Detected language of conversation
wa_conversations	message_count	integer	default 0	Total messages in this conversation
wa_conversations	last_message_at	timestamptz		Timestamp of last message
wa_conversations	escalated	boolean	default false	Whether conversation was escalated to human
wa_messages				
wa_messages	id	uuid	PK, default gen_random_uuid()	Message ID
wa_messages	created_at	timestamptz	NOT NULL, default now()	Message timestamp
wa_messages	conversation_id	uuid	FK → wa_conversations.id	Parent conversation
wa_messages	direction	text	CHECK (direction IN ('inbound','outbound'))	Message direction
wa_messages	body	text	NOT NULL	Message text content
wa_messages	language	text		Detected language of this message
wa_messages	claude_model	text		Claude model used for response (outbound only)
wa_messages	response_time_ms	integer		Time from receipt to response sent (outbound only)
wa_messages	twilio_sid	text		Twilio message SID for tracking
wa_broadcasts				
wa_broadcasts	id	uuid	PK, default gen_random_uuid()	Broadcast ID
wa_broadcasts	created_at	timestamptz	NOT NULL, default now()	Broadcast timestamp
wa_broadcasts	template_name	text	NOT NULL	Twilio/Meta template name used
wa_broadcasts	content_fr	text		French message content
wa_broadcasts	content_en	text		English message content
wa_broadcasts	target_group	text		Target group: all, vip, specific channel
wa_broadcasts	recipients_count	integer	default 0	Number of recipients
wa_broadcasts	delivered_count	integer	default 0	Number of confirmed deliveries
wa_broadcasts	sent_by	text		Admin who triggered the broadcast
wa_analytics_daily				
wa_analytics_daily	id	uuid	PK, default gen_random_uuid()	Row ID
wa_analytics_daily	date	date	NOT NULL, UNIQUE	Date
wa_analytics_daily	total_messages	integer	default 0	Total messages received
wa_analytics_daily	unique_users	integer	default 0	Unique sender hashes
wa_analytics_daily	avg_response_ms	integer		Average response time in milliseconds
wa_analytics_daily	escalations	integer	default 0	Number of escalations to human
wa_analytics_daily	lang_fr	integer	default 0	Messages in French
wa_analytics_daily	lang_en	integer	default 0	Messages in English
wa_analytics_daily	lang_pt	integer	default 0	Messages in Portuguese
wa_analytics_daily	lang_other	integer	default 0	Messages in other languages

Reference Documentation & Links

Resource	URL	Relevance
Twilio WhatsApp Overview	https://www.twilio.com/docs/whatsapp/api	Primary Twilio docs
Twilio Self Sign-up Guide	https://www.twilio.com/docs/whatsapp/self-sign-up	Sender registration process
Twilio Key Concepts	https://www.twilio.com/docs/whatsapp/key-concepts	24h windows, templates, BSUIDs
Twilio Quickstart	https://www.twilio.com/docs/whatsapp/quickstart	First message tutorial
Twilio Best Practices & FAQs	https://www.twilio.com/docs/whatsapp/best-practices-and-faqs	Common issues and solutions
Twilio WhatsApp Business Accounts	https://www.twilio.com/docs/whatsapp/tutorial/whatsapp-business-account	WABA setup, display names, OBA
Twilio Tutorials & Sample Code	https://www.twilio.com/docs/whatsapp/tutorial	Code examples
World Bank WhatsApp-RAG-Example	https://github.com/worldbank/WhatsApp-RAG-Example	Reference architecture: FastAPI + Twilio + LLM
Anthropic Claude API Docs	https://docs.anthropic.com/	Claude API reference
Supabase Docs	https://supabase.com/docs	Database, auth, storage
FastAPI Documentation	https://fastapi.tiangolo.com	Backend framework
OAFLAD Official Website	https://oaflad.org/en/	OAFLAD branding, content reference
Meta Business Verification	https://www.facebook.com/business/help/1095661473946872	Meta verification process